

UniReflex: Plug-and-Play Force Control for Pretrained Generative Policies via Fast-Slow Reflex

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Abstract—Generative imitation learning policies excel at trajectory planning but lack closed-loop force regulation, while directly incorporating force modalities often requires redesigning or retraining the network. We present UniReflex, a universal plug-and-play framework that equips frozen generative policies with variable impedance control (VIC) for contact regulation, guided by force-direction intent collected during demonstration, without further slow-backbone fine-tuning. By non-invasively intercepting deep latent representations from the action head, UniReflex drives a fast reflex network that decouples active force exertion from external interaction response. This scheme predicts normalized anisotropic stiffness directions for directional compliance allocation. Furthermore, UniReflex integrates an adaptive gating mechanism that enables seamless transitions between position-dominant planning and force-dominant execution. Real-world bimanual experiments demonstrate that UniReflex significantly improves contact stability and success rates while preserving original position accuracy. Our approach achieves $25\text{--}66\times$ lower per-step backward latency relative to joint training strategies on the evaluated backbones. Project website: <https://unireflex.github.io/>.

Index Terms—Force Control, Compliance and Impedance Control, Imitation Learning

I. INTRODUCTION

Imitation learning has become the dominant recipe for scalable visuomotor manipulation: Diffusion Policy (DP) [1], vision-language-action (VLA) models [2], and world action models (WAM) that learn predictive dynamics or video-grounded latents before decoding actions [3] now reach strong performance on diverse tabletop and pick-and-place regimes. Yet contact-rich, force-sensitive skills, such as wiping, insertion, peeling, and sustained pressing, often remain unreliable in deployed systems. The dominant recipe predicts kinematic motion targets at the policy level and tracks them with approximately fixed low-level gains; under this pattern, small variations in geometry, compliance, or disturbance can destabilize contact in ways that are difficult to remedy through data scaling alone. This fundamental limitation highlights a gap between purely geometric motion planning and the physical realities of robotic deployment.

In parallel, a growing line of work targets contact by making forces explicit in learning and control, e.g., force-conditioned reactive policies [4], force-aware VLAs [5], hybrid force-position formulations [6], and tactile or vision-driven fast-slow policies [7]. These advances show that wrench and tactile cues can materially improve manipulation, but they also diversify how force information is used. By directly incorporating physical interaction data, these methods



Fig. 1. **Overview of UniReflex.** A general plug-and-play force-control framework with strong disturbance rejection and force tracking.

attempt to bridge the gap between visual intent and tactile execution.

Concretely, recent methods cluster into three families. First, many approaches treat force or torque primarily as additional observations: force/torque or tactile streams are fused into the policy input while the network still predicts trajectory-like actions [4], [5]. Second, another family couples visuomotor policies with hybrid force-position or variable impedance control (VIC), explicitly regulating some degrees of freedom in force and others in pose [6], [8]. Third, fast-slow hierarchies decouple slow visual reasoning from high-rate contact feedback [7], e.g., via tactile reactive heads or cascaded decoders [11]. Across these families, however, a recurring practical cost is structural: realizing their full pipelines typically requires end-to-end training, architectural refits, or joint optimization. This poses a significant dilemma in the era of foundation models. Today’s most capable generative policies are pre-trained on massive, predominantly vision-only datasets that lack explicit force information. Discarding these powerful pre-trained weights—or subjecting them to heavy fine-tuning to accommodate new force modalities—is highly inefficient and risks destroying their well-generalized distributions. Because force feedback is a specialized and relatively sparse observation, it should ideally be integrated as an independent module. A principled solution must therefore preserve the original pre-trained distribution of the foundation model while independently endowing the system with robust physical reflexes.

This raises a direct question: can we add closed-loop contact regulation as a **lightweight, plug-and-play module** without further fine-tuning the already task-adapted slow policy, while still benefiting from its high-level intent? We present **UniReflex** (Fig. 1), a general plug-and-play force-

TABLE I

TAXONOMY OF FORCE CONTROL METHODS. UNIREFLEX UNIQUELY ENABLES ACTIVE, VARIABLE-STIFFNESS CONTROL WHILE MAINTAINING PLUG-AND-PLAY COMPATIBILITY WITH FROZEN GENERATIVE POLICIES VIA FAST-ONLY TRAINING.

Category	Representative methods	Force ctrl.	Closed-loop	Variable stiff.	Ref. force	Plug-play	Fast-only training
Variable impedance control							
Guided Control	ACP [8]	Active	✓	✓	✓	✗	✗
Hybrid P/F Control	ForceVLA2 [9]	Active	✓	✓	✓	✗	✗
Force-conditioned imitation learning							
Force-aware DP	FoAR [4]	Passive	✗	✗	✗	✗	✗
Force-aware VLA	TA-VLA [10], ForceVLA [5]	Passive	✗	✗	✗	✗	✗
Fast-slow systems							
Tactile-reactive DP	RDP [7]	Passive	✓	✗	✗	✗	✗
VLA-diffusion hybrid	RoboDual [11]	Passive	✗	✗	✗	✓	✓
UniReflex (Ours)	-	Active	✓	✓	✓	✓	✓

control framework that attaches fast contact regulation to frozen generative policies through a decoupled design for denoising/generative backbones: no model-specific fast-slow interface is required, and only the lightweight reflex is trained while the backbone stays frozen. We further verify that directly reusing the latent action representation from the action generation head provides a recoverable representation for downstream force control, and that the same reflex interface transfers across heterogeneous generative policies. By doing so, we aim to offer a scalable alternative to monolithic end-to-end training pipelines. Table I provides a compact, deployment-oriented comparison of representative methods. Our contributions are threefold:

- 1) **A Universal Plug-and-Play Framework:** UniReflex equips supervised fine-tuned generative policies with closed-loop force control by non-invasively intercepting deep action-head latent representations, without additional training or fine-tuning of the slow backbone.
- 2) **Force-Stiffness Decoupling:** We jointly supervise normalized anisotropic stiffness directions and reference forces, enabling VIC-based closed-loop regulation that yields more physically consistent wrench behavior.
- 3) **Adaptive Gating for Hybrid Control:** A lightweight gating mechanism switches between position-dominant and force-dominant execution so the robot preserves the frozen planner’s alignment quality while gaining robust regulation during contact.

II. RELATED WORK

We organize concurrent work along the same axes as the Introduction: high-capacity imitation policies, force-aware learning and control, and fast-slow decompositions. Table I provides a compact architectural comparison based on strict operational criteria: **Active** force control explicitly commands force/stiffness targets, whereas **Passive** relies purely on kinematics; **Closed-loop** denotes real-time action adjustment specifically via force, torque, current, or tactile

feedback; **Variable stiff.** and **Ref. force** indicate explicit prediction of stiffness parameters and desired force trajectories; **Plug-play** and **Fast-only** highlight the ability to avoid fine-tuning the frozen slow backbone. Here we summarize each thread and how UniReflex differs in problem setting.

A. Generative Imitation Learning for Manipulation

Modern generative imitation policies, including diffusion trajectory models [1], VLAs [2], and WAM [3], achieve strong visuomotor performance on diverse manipulation benchmarks. Efficient fine-tuning recipes remain the default path for adapting large VLAs to new robots, and dual-encoder strategies that preserve frozen pretrained vision-language representations while adapting robot-specific pathways have been proposed to mitigate catastrophic forgetting during VLA adaptation. Their standard interfaces nevertheless remain dominated by kinematic action prediction with fixed-gain tracking, so contact wrenches are rarely regulated as explicit closed-loop objectives at deployment.

B. Force-Aware Robot Control

Classical and learning-based impedance and variable-stiffness control [12], [13], [14] shape compliant contact by modulating stiffness and damping over time. Recent learning systems further couple visuomotor policies with hybrid force-position execution [8], [5], [6], [9], [15]. In parallel, force-conditioned imitation learning incorporates force/torque or tactile streams as observations [4], [16], [10], [5], improving contact-rich behavior through representation learning even when the deployed controller is not a classical hybrid force-position controller. Across these families, realizing their full pipelines typically requires end-to-end training or coordinated fine-tuning of perception and action, rather than composing a post-hoc module atop an already trained generative policy.

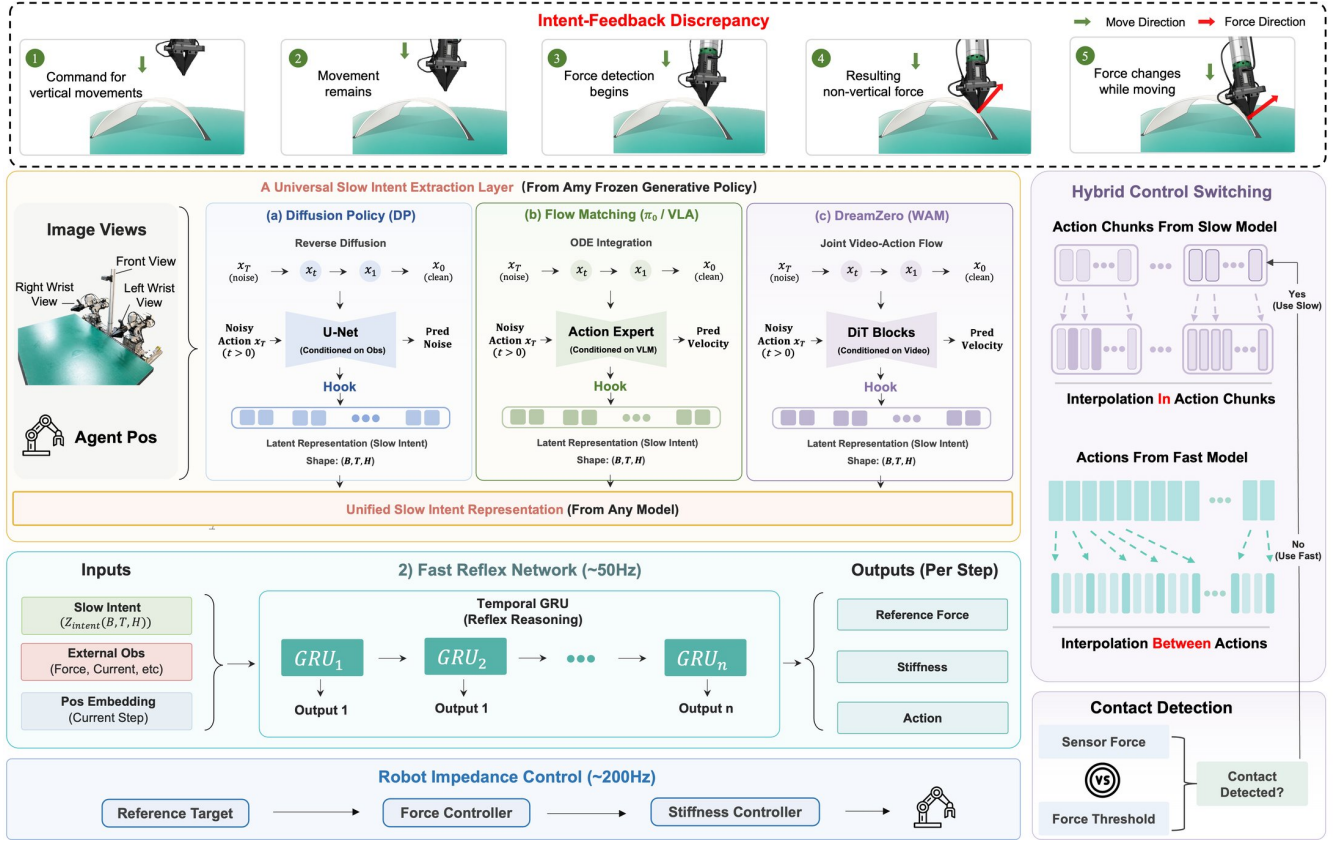


Fig. 2. **System architecture of UniReflex.** UniReflex hierarchically couples slow vision-driven planning, a fast proprioceptive reflex, and a real-time execution layer that tracks predicted reference forces together with reference-impedance targets.

C. Hierarchical Fast-Slow Policies

Fast-slow and dual-rate hierarchies address the latency gap between slow VLA inference and millisecond-scale contact by cascading decoders, tactile heads, or asynchronous buffers [7], [11]. These designs improve temporal alignment for contact, but prior art typically *co-trains* slow and fast modules, which is expensive for large backbones and misaligned with upgrading frozen, field-deployed policies. FGIL has explored generating joint-level force commands from a provided kinematic trajectory with a memoryless network and an external PID loop [17]. However, it assumes that trajectory is already available, typically from direct teaching, rather than attaching reference-force and variable-stiffness control to a frozen visuomotor generative policy. UniReflex instead keeps the slow generative policy frozen, trains only a compact reflex on proprioceptive feedback and hooked action-head features, and targets active variable-stiffness wrench regulation.

III. PRELIMINARIES

A. Generative Action Heads

Modern generative policies differ in objectives and architectures: DP [1] predicts noise (ϵ), while VLAs like π_0 [2] and WAMs like DreamZero [3] predict flow velocities (v_t). Despite this, their action heads share a core pattern: decoding actions via iterative denoising or flow integration. We unify

these paradigms at their final generation step ($k \rightarrow 0$). Here, the internal state closely approximates the clean action sequence x_0 . To predict the final tiny residual or velocity, the network must encode a comprehensive trajectory understanding within its hidden layers. Before the final projection collapses this into low-dimensional kinematics, models maintain a temporally structured hidden representation. We reshape this pre-projection state as $z \in \mathbb{R}^{B \times T_{\text{slow}} \times H}$ (batch, action horizon, hidden width). UniReflex intercepts this as a backend-agnostic “slow intent” interface, preserving task semantics without architectural surgery or slow-policy fine-tuning.

B. Variable Impedance Control

At execution time, contact-rich manipulation is often closed with variable impedance control (VIC): the end-effector is regulated as a *virtual* mass–spring–damper instead of as a stiff pose tracker. In Cartesian coordinates, a standard error-based model is

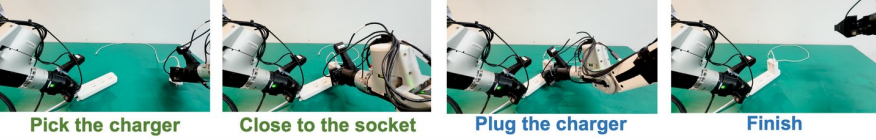
$$\Lambda \ddot{\tilde{x}} + D(t) \dot{\tilde{x}} + K(t) \tilde{x} = \mathbf{F}_{\text{ext}}, \quad (1)$$

where $\tilde{x}$ is end-effector pose error, $\mathbf{F}_{\text{ext}}$ is the contact wrench measured at the wrist, $K(t)$ sets how stiff the arm is along each axis, and Λ fixes the apparent end-effector inertia. In practice, the time-varying gains $K(t)$ and $D(t)$ are commonly scheduled online from contact measurements together with task-phase logic.

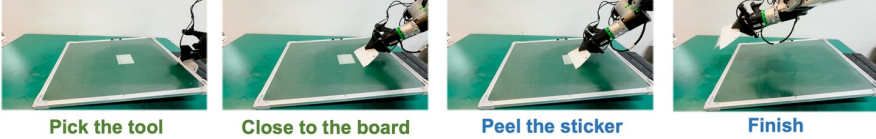
Flip Chip Box: Flip the chip box to a vertical position.



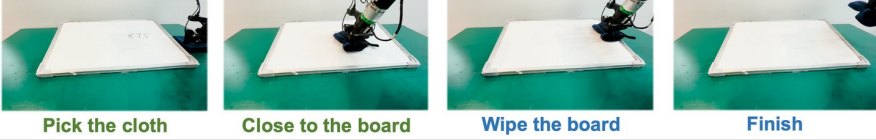
Plug Charger: Insert the plug into the power socket.



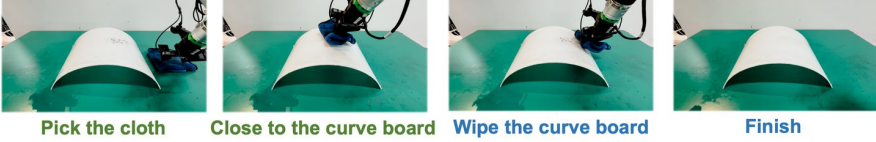
Peel Note: Peel the sticky note off the surface.



Wipe Incline: Remove marks from the **inclined whiteboard** using a cloth.



Wipe Curve: Remove marks from the **curved surface** using a cloth.



Position Control

Force Control

Unstable bottom



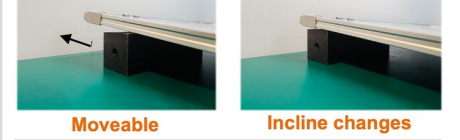
Unfixed socket



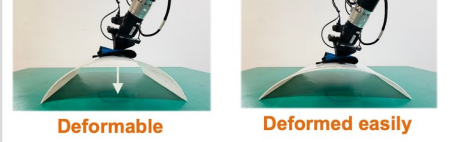
Different incline angles



Different incline angles



Prone to deformation



Main Challenges

Fig. 3. **Five contact-rich tasks:** short per-task introductions and the main physical challenges, together with a two-stage evaluation protocol.

IV. METHOD

Fig. 2 overviews the resulting hierarchy: a frozen slow generative policy, a fast proprioceptive reflex, and a VIC execution layer.

A. Active Effort Recovery and Variable-Stiffness Labeling

We build supervision signals from teleoperation logs. Consider an n -degree-of-freedom bilateral teleoperation stack whose master and slave arms obey

$$M_m(q_m)\ddot{q}_m + C_m(q_m, \dot{q}_m)\dot{q}_m + G_m(q_m) = \tau_m + \tau_h, \quad (2)$$

$$M_s(q_s)\ddot{q}_s + C_s(q_s, \dot{q}_s)\dot{q}_s + G_s(q_s) = \tau_s + \tau_e, \quad (3)$$

where (q_m, q_s) are joint coordinates, (τ_h, τ_e) are interaction torques, and quasi-static motion with stiff slave tracking $\tau_s = K_p(q_m - q_s) + K_d(\dot{q}_m - \dot{q}_s) + G_s(q_s)$ implies $\tau_e \approx K_p(q_m - q_s)$; mapping this relation to Cartesian end-effector coordinates yields a loggable proxy for interaction-force intent,

$$F_{\text{intent}} \approx K_{\text{virtual}} \Delta x, \quad \Delta x = x_m - x_s, \quad (4)$$

where K_{virtual} is the effective Cartesian stiffness map implied by the slave position loop and the kinematic Jacobian.

In physical interaction, stiffness should rise with commanded effort rather than relying on stiffness reduction alone for compliance. We apply this qualitative rule to demonstrations: stronger mismatch cues along the teleoperation error direction justify a stiffer axis in the supervised impedance targets, improving contact stability with wrist force/torque sensing only (no extra hybrid force-position hardware).

Let $n = \Delta x / \|\Delta x\| \in \mathbb{R}^3$ be the principal effort axis from teleoperation mismatch. To ensure stability on standard industrial controllers, we project this intent into a decoupled diagonal stiffness template,

$$K_{\text{label}} = \text{diag}(|n_x|, |n_y|, |n_z|) + \epsilon I, \quad \epsilon > 0, \quad (5)$$

which encodes anisotropic stiffness magnitudes while simplifying the learning objective and guaranteeing positive-definiteness.

B. Non-Invasive Internal Representation Interception

Building on the shared action-head structure in Section III-A, we read multimodal intent from *late* internal activations without rewiring the vision stack. Crucially, we observe that whether the generative mechanism is diffusion (DP) or flow matching (VLA and WAM), the policy invariably forms a

TABLE II
SUCCESS RATES (%) ON CONTACT-RICH BENCHMARKS. PHASE I VS. PHASE II; UNIREFLEX-AUGMENTED POLICIES VS. VANILLA BACKBONES AND FORCE-AWARE BASELINES.

Policy	Wipe (Curve)		Wipe (Incline)		Plug Charger		Peel Note		Flip Chip Box	
	lift	wipe	lift	wipe	lift	plug	lift	peel	push	flip
RDP	60.0%	50.0%	80.0%	70.0%	60.0%	50.0%	60.0%	50.0%	70.0%	50.0%
ForceVLA	30.0%	20.0%	50.0%	30.0%	30.0%	0.0%	50.0%	30.0%	40.0%	20.0%
TA-VLA	50.0%	50.0%	70.0%	50.0%	40.0%	40.0%	70.0%	40.0%	50.0%	30.0%
DP	70.0%	30.0%	80.0%	30.0%	70.0%	50.0%	90.0%	30.0%	80.0%	30.0%
π_0	80.0%	30.0%	90.0%	30.0%	80.0%	50.0%	90.0%	30.0%	90.0%	30.0%
DreamZero	60.0%	20.0%	60.0%	20.0%	50.0%	30.0%	70.0%	10.0%	70.0%	20.0%
UniReflex + DP	80.0%	70.0%	80.0%	70.0%	80.0%	50.0%	90.0%	90.0%	80.0%	70.0%
UniReflex + π_0	80.0%	80.0%	90.0%	80.0%	70.0%	50.0%	90.0%	80.0%	80.0%	80.0%
UniReflex + DreamZero	60.0%	50.0%	70.0%	50.0%	60.0%	30.0%	60.0%	30.0%	60.0%	50.0%

temporally structured hidden representation prior to final action projection. We formalize this intercepted feature as a unified latent representation, which encapsulates the slow policy’s intent:

$$z = \text{Hook}(\pi_{\text{gen}}, o_t) \in \mathbb{R}^{B \times T_{\text{slow}} \times H}, \quad (6)$$

where B is the batch size, T_{slow} is the action horizon, and H is the hidden dimension.

Concretely, the noise (or flow) predictor admits a standard decomposition,

$$\epsilon_\theta(x_k, o_t, k) = W_{\text{out}} \Phi_{\text{dec}}(x_k, o_t, k), \quad (7)$$

where Φ_{dec} aggregates deep multimodal reasoning and W_{out} maps to action coordinates. Near the end of reverse synthesis ($k \rightarrow 0$), the sampled trajectory geometry is largely committed; we register a *forward* hook on Φ_{dec} to extract z . The hook is read-only with respect to frozen backbone weights, so the slow policy needs neither architectural surgery nor joint fine-tuning. This unified $\mathbb{R}^{B \times T_{\text{slow}} \times H}$ interface allows UniReflex to act as a backend-agnostic module across heterogeneous generative policies.

C. Intent-Conditioned State Observation and Reflex Decoding

Many generative policies use action chunking, predicting a T_{slow} -step horizon $A_{\text{chunk}} = [x_t, \dots, x_{t+T_{\text{slow}}-1}]$ in one pass; under contact, open-loop rollout of long chunks reacts poorly to impacts and friction. The reflex ϕ_{reflex} therefore emits a single-step update each control cycle; we employ a causal temporal model so short-term contact dynamics are retained in memory. At time t , the fast network receives real-time extended observations e_t and the unified latent representation $z \in \mathbb{R}^{B \times T_{\text{slow}} \times H}$ intercepted from π_{gen} , which serves as the slow intent.

To bridge the temporal gap between the slow chunk and the high-frequency reflex, we first temporally align the latent representation z to the fast action horizon, yielding z'_t . At each fast step t , the input u_t is constructed by concatenating three encoded features: the step-wise slow intent condition

$s_t = f_{\text{slow}}(z'_t)$, the real-time feedback $r_t = f_{\text{ext}}(e_t)$, and a learnable step positional embedding $p_t = \text{Emb}(t)$ that indicates the current phase within the chunk,

$$u_t = [f_{\text{slow}}(z'_t) \oplus f_{\text{ext}}(e_t) \oplus \text{Emb}(t)]. \quad (8)$$

Given an initial hidden state h_0 derived from the latent representation and the initial agent pose, the internal state then evolves via a causal temporal model (instantiated as a GRU in our implementation, though compatible with other causal sequence encoders),

$$h_t = \mathcal{F}_\theta(u_t, h_{t-1}). \quad (9)$$

Finally, a generic decoder function $\mathcal{G}_\phi(\cdot)$ maps the internal state h_t along with the current input u_t to three execution primitives,

$$\begin{bmatrix} x_{\text{reflex}} \\ F_{\text{ref}} \\ \Phi_K \end{bmatrix} = \mathcal{G}_\phi(h_t, u_t). \quad (10)$$

The decoder emits three coupled outputs: x_{reflex} matches the backbone pose format and, streamed each frame, can override the chunk after contact to limit geometric drift; F_{ref} is the desired contact wrench for sustained normal-pressure tracking; and Φ_K encodes end-effector stiffness and supports decoupling the direction and magnitude of applied effort.

D. Contact-Triggered Phase Gating

At deployment we alternate between two execution regimes consistent with our benchmarks. **Phase I:** before sustained contact, the low-level controller prioritizes tracking the frozen generative policy’s pose targets. **Phase II:** once contact is detected from proprioceptive sensing, we switch emphasis to regulating F_{ref} and the impedance parameters implied by Φ_K through the VIC interface, while still streaming single-step x_{reflex} corrections for geometric drift.

V. EXPERIMENTS

We design experiments to answer six questions: **(Q1)** Can UniReflex improve contact-stage performance while preserving the backbone’s non-contact visuomotor behavior?

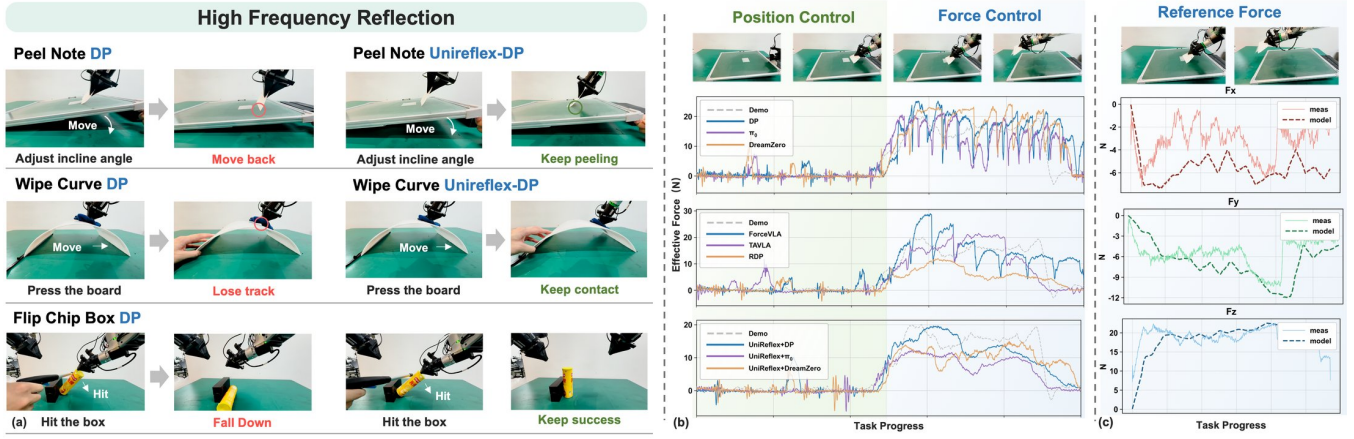


Fig. 4. **High-frequency reflex and force tracking:** (a) UniReflex maintains success on contact-rich tasks under disturbances. (b) Effective force during the peel task for baselines vs. UniReflex. (c) Measured contact force vs. predicted reference force during UniReflex execution.

(Q2) How does UniReflex compare with end-to-end force-aware policies in position- and force-dominated execution? (Q3) Do learned force/stiffness outputs, high-frequency residuals, and contact-aware gating account for these gains? (Q4) Does the fast sub-network enable high-frequency recovery under environmental disturbances? (Q5) Does UniReflex improve force tracking and contact robustness during interaction? (Q6) Does its lightweight reflex design provide superior training throughput and memory efficiency over joint optimization?

A. Setup

We evaluate five contact-rich benchmarks (Fig. 3) under the two-phase protocol of Section IV-D: Phase I scores approach or alignment (lift/push), and Phase II scores the subsequent contact skill (wipe/peel/flip/plug). Our primary study couples UniReflex with three mainstream backbones: **DP** [1], π_0 [2], and **DreamZero** [3]. In all cases, the generative stack remains frozen (i.e., already task-adapted) and only UniReflex is trained (Stage II) in a plug-and-play manner without further slow-policy fine-tuning. We additionally report force-aware monolithic baselines RDP [7], ForceVLA [5], and TA-VLA [10] (Table II). Demonstrations are collected via teleoperation at 200 trajectories per task, and each reported cell is evaluated over 10 rollouts.

B. Results

Q1: Task Performance Across Execution Phases. For each backbone $X \in \{\text{DP}, \pi_0, \text{DreamZero}\}$, Table II compares UniReflex+ X with vanilla X . We report task-level success over the two consecutive phases: Phase II is attempted only after successful completion of Phase I, and a Phase I failure is therefore retained as a failure in the Phase II success rate. This end-to-end measure accounts for the approach and alignment behavior required before force-regulated execution can begin. Across all tasks, adding UniReflex changes Phase I success by no more than 10 percentage points relative to the corresponding backbone, indicating that visuomotor performance during the non-contact

phase is largely preserved. In Phase II, UniReflex improves success by 20–60 percentage points on every sustained-contact benchmark across all three backbones, e.g., Peel Note with DP (30.0% to 90.0%) and Flip Chip Box with π_0 (30.0% to 80.0%). DreamZero also gains 20–40 points but remains below the DP/ π_0 pair. Plug Charger is the only task without a clear improvement (the plug cell is unchanged at 50.0%/50.0%/30.0%). Unlike the other benchmarks, it is dominated by brief contact transients and tight insertion tolerances, which leave less scope for sustained force regulation.

Q2: Force-Dominated Execution vs. Monolithic Baselines. Table II contrasts UniReflex+DP [1], UniReflex+ π_0 [2], and UniReflex+DreamZero [3] with monolithic RDP [7], ForceVLA [5], and TA-VLA [10]. On Phase I, the frozen planner keeps the original backbone’s position-control quality: UniReflex+DP matches or improves every alignment cell relative to vanilla DP, whereas the end-to-end force-aware models degrade on the same columns. ForceVLA and TA-VLA retrain the full stack around wrench or tactile observations and therefore trade away kinematic alignment, often trailing by tens of points; RDP is lower or tied on each lift/push slot. UniReflex+ π_0 also outscores all three monoliths on every Phase I cell. On Phase II, UniReflex+DP and UniReflex+ π_0 meet or beat the best monolithic numbers (e.g., Peel Note 90.0%/80.0% versus RDP 50.0%). RDP is the strongest force-aware baseline because it already closes a tactile loop, but it remains a passive kinematic policy and still sits below our DP and π_0 entries on most contact-heavy cells. UniReflex+DreamZero is closer to RDP than to the other two UniReflex variants, consistent with its slower chunk refresh.

Q3: Component Contributions. As shown in Table III, we conduct two ablation studies with a frozen DP backbone on Peel Note, Wipe (Curve), and Wipe (Incline) to explain the gains observed in Q1.

We test four alternative force/stiffness configurations: (i) disabling learned stiffness modulation while retaining the baseline stiffness; (ii) replacing the learned dynamic

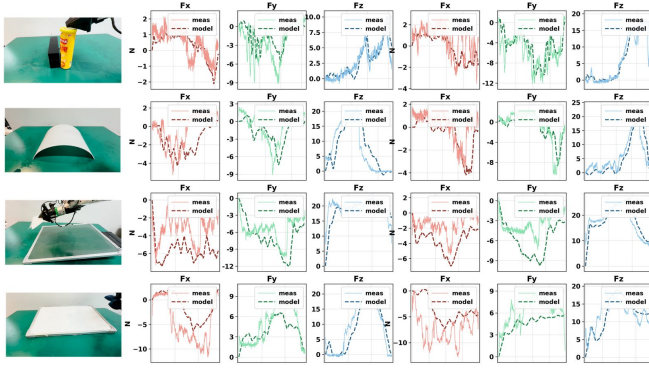


Fig. 5. **Force tracking across representative tasks.** Each row pairs a task snapshot with measured (solid) and model (dashed) F_x , F_y , and F_z traces. The measured wrench tracks the predicted reference through contact, with residual deviations corrected at the reflex rate.

anisotropic stiffness with a manually selected isotropic stiffness that remains fixed during each task; (iii) retaining only reference-wrench prediction while removing the stiffness-direction output (wrench-only); and (iv) replacing state-dependent learned stiffness with a hand-designed controller using per-task fixed impedance schedules and manually calibrated gains. Alongside Phase II success, we quantify contact-stage force tracking using

$$F\text{-err.} = \frac{\sum_{t=1}^{T_{II}} \|F_t^{\text{meas}} - F_t^{\text{tar}}\|_1}{\sum_{t=1}^{T_{II}} \|F_t^{\text{tar}}\|_1}, \quad (11)$$

where T_{II} is the number of Phase II samples and F_t^{meas} and F_t^{tar} are the measured and target force vectors; F-err. is reported as a percentage. Across the first three component-reduced configurations, Phase II success drops from 70.0–90.0% to 20.0–50.0%, while F-err. rises from 5.8–8.9% to 28.3–38.7%. The wrench-only variant is weakest, so a reference force without directional stiffness is not enough. The hand-designed controller is better (F-err. 17.6–19.2%) but still below the learned formulation.

The second study separates stiffness modulation from contact gating. The gated Δx GRU retains the gate but replaces learned stiffness with Δx -based force regulation: it preserves 80.0–90.0% Phase I success, yet pose error accumulates and Phase II drops to 40.0–60.0% versus 70.0–90.0% for full UniReflex. The ungated GRU uses the fast network in both phases and collapses to 0.0–30.0%/0.0–10.0%. Thus the learned wrench–stiffness pair is not interchangeable with a residual pose servo, and gating is required to protect the frozen planner before contact. This also explains the Q1 pattern: Phase I stays close to the backbone whenever the gate is on, while Phase II rises only when both wrench and stiffness outputs remain.

Q4: High-Frequency Recovery under Disturbances.

To evaluate rapid reactions to online workspace changes, we test three tasks under dynamic disturbances, as shown in Fig. 4. Specifically, we continuously adjust the support inclination during Peel Note, press the curved board during Wipe (Curve) to induce deformation, and perturb the chip box during Flip Chip Box to create unexpected contact

TABLE III
COMPONENT ABLATIONS UNDER FROZEN DP.

Variant	Peel Note	Wipe (Curve)	Wipe (Incline)
Force/stiffness output (Phase II success / F-err.)			
UniReflex (DP)	90.0% / 5.8%	70.0% / 8.9%	70.0% / 6.8%
No stiffness mod.	50.0% / 32.4%	30.0% / 38.7%	40.0% / 35.6%
Fixed isotropic stiffness	50.0% / 28.3%	40.0% / 34.2%	30.0% / 36.5%
Wrench-only	30.0% / 34.1%	20.0% / 37.4%	20.0% / 33.8%
Manual impedance	60.0% / 18.4%	50.0% / 19.2%	40.0% / 17.6%
Residual/gating controls (Phase I / Phase II success)			
UniReflex (DP)	90.0% / 90.0%	80.0% / 70.0%	80.0% / 70.0%
Gated Δx GRU	90.0% / 60.0%	80.0% / 40.0%	80.0% / 50.0%
Ungated GRU	0.0% / 0.0%	30.0% / 0.0%	30.0% / 10.0%

TABLE IV
RECOVERY UNDER TWO PERTURBATION SETTINGS.

Perturbation	Policy	Metric	Peel	Wipe	Flip
Dynamic	DP	Rec.	10.0%	20.0%	10.0%
	UniReflex (DP)	Rec./ T_{rec}	90.0%/0.87s	100.0%/1.13s	80.0%/1.28s
TCP bias (1–2 cm)	UniReflex (DP)	Phase II Rec.	80.0%	90.0%	70.0%

changes. As shown in Fig. 4(a), frozen DP loses contact under disturbances, whereas UniReflex+DP maintains reflex corrections and completes the tasks.

Table IV reports 10 trials per task under two perturbations. Recovery is the fraction of trials that re-establish the commanded contact and finish Phase II; T_{rec} is the time from disturbance onset until the measured force re-enters the target band. Under dynamic disturbances, UniReflex recovers 80.0–100.0% of executions within 0.87–1.28s, versus 10.0–20.0% for frozen DP, which continues the stale open-loop chunk. With a 1–2 cm pre-contact TCP bias, UniReflex still recovers Phase II in 70.0–90.0% of trials.

Q5: Interaction-Time Force Tracking. As shown in Fig. 4(b), UniReflex-augmented policies track demonstration force trajectories much more closely during the peeling task than frozen baselines. This stability explains their higher Phase II success in contact-rich tasks, especially when visual cues are sparse and force control is required to stabilize interaction. Focusing on DP (Fig. 4(c)), UniReflex sustains peel contact by aligning measured 3D forces (F_x , F_y , F_z) with the predicted references, whereas frozen DP fails to keep forces within the contact-compatible band. Fig. 5 extends this comparison across four representative tasks. Each row pairs a workspace snapshot with measured (solid) and model (dashed) F_x , F_y , and F_z traces. As contact conditions change across each task, the measured wrench continues to track this time-varying predicted reference instead of locking onto a single constant force. Deviations remain visible, as expected under friction and geometric change, but the reflex loop continuously corrects them at the fast control rate rather than waiting for the next slow-chunk update.

Q6: Training Throughput and Memory Efficiency. As shown in Table V, we compare estimated joint fine-tuning of the full stack with UniReflex Stage II, which trains only the fast GRU, on a single NVIDIA A800 GPU. Across the

TABLE V

BACKBONE SCALE VS. REFLEX TRAINING COST. JOINT FT: ESTIMATED FULL FORWARD-BACKWARD COST; **OURS:** FAST GRU BACKWARD STEP ONLY.

Backbone	Slow/Fast Params	Ratio	Steps K / Size H	Joint FT (ms)	Ours (ms)	Speedup
DP	452.3M / 1.70M	0.375%	20 (DDIM) / 256	≈ 406	7.70	$\approx 53\times$
π_0	3.50B / 6.60M	0.188%	10 (Flow-Matching) / 1,024	≈ 566	8.57	$\approx 66\times$
DreamZero	22.92B / 62.2M	0.271%	8 (Flow-Matching) / 5,120	$\approx 5,490$	223	$\approx 25\times$

evaluated backbones, this reduces backward-pass latency by $25\text{--}66\times$ while training only 0.188–0.375% as many parameters as the backbone. Joint-FT cost grows with backbone size and sampling steps K , whereas the UniReflex backward step tracks only the GRU width H ; DreamZero is therefore slower in absolute milliseconds ($H = 5120$) but still about $25\times$ cheaper than updating 22.9B frozen weights. The smaller trainable state also reduces optimizer-state storage, so a new backbone can keep its pretrained weights and only the reflex is optimized. In short, the training budget of UniReflex scales with the reflex, not with the frozen generative policy. This is the main reason a 3.5B or 22.9B backbone can be reused as-is: Stage II never backpropagates through the slow denoising or flow stack, so adding force control does not require a second full pretraining or fine-tuning run.

VI. LIMITATIONS

First, gains are smaller for long-horizon world-action backbones such as DreamZero, because their slowly refreshed action chunks limit timely responses to physical contact changes; this is already visible as a lower Phase II ceiling in Table II. Second, UniReflex is better suited to sustained contact than brief, high-impulse rigid insertion, making tasks such as Plug Charger more challenging. Third, achievable responsiveness remains bounded by force-sensing quality, low-level control, and hardware bandwidth; the reflex assumes a wrist force/torque signal and a VIC interface, and cannot create contact information that the robot does not sense. These limits are orthogonal to the plug-and-play training recipe: they bound execution quality even when the slow backbone remains frozen.

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